

A FULL-STACK SOLUTION FOR FUTURE
CS PROFESSIONALS

Designing an Academic and Career Development Website for Computer Science Students

presentation



Team : 02

Members:

Zhang Jieding DC228922

Yang Yujie DC229823

Chen Siyi DC228141

Zhang Zhexuan DC229576

Luo Yiming DC229226

Yang Hao DC229340



01

Target Audience & Project Purpose



Target Audience



Core Purpose





PROBLEM AND SOLUTION



- Problem: career and academic planning is overwhelming
- Information is scattered
- Students make decisions too late
- No clear structure for comparing paths
- Our solution: one structured full-stack platform



TECHNICAL APPROACH

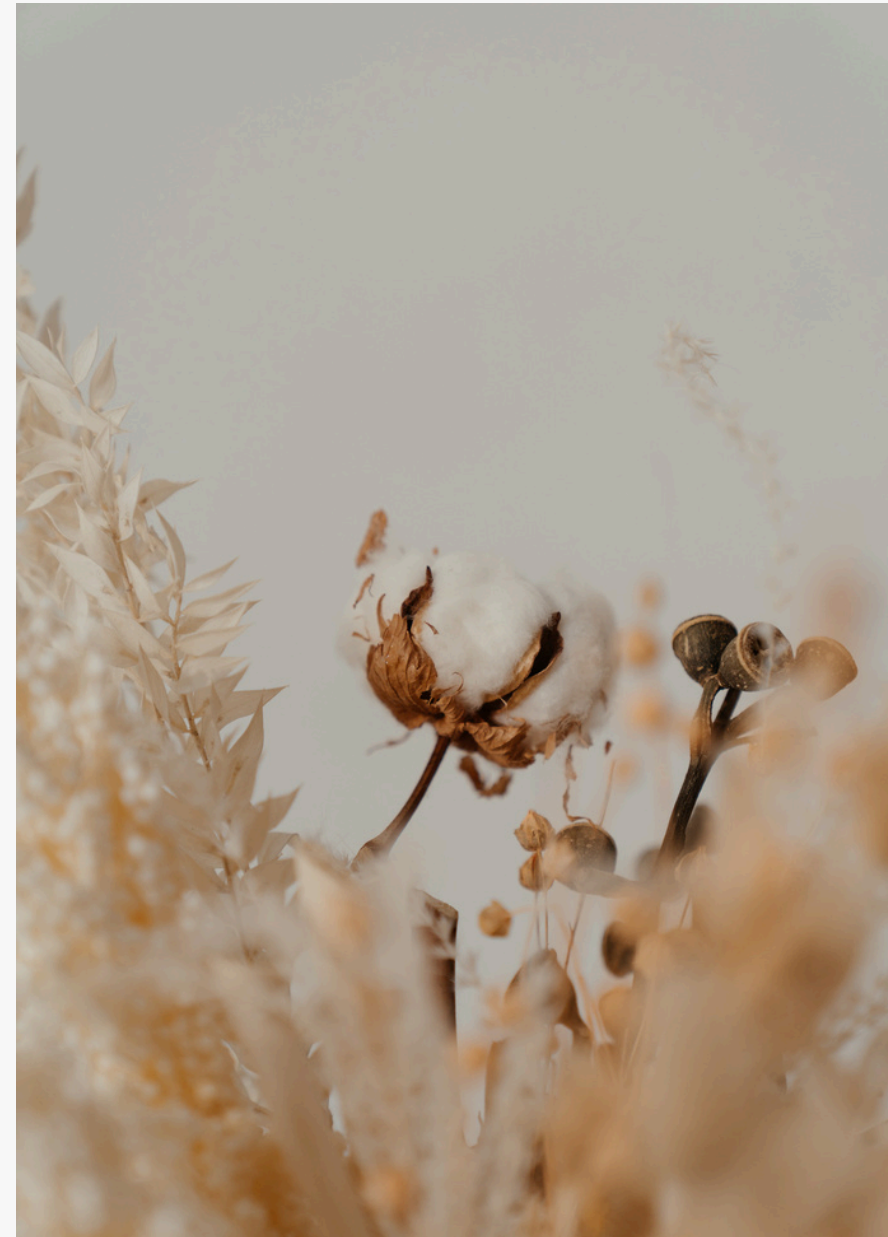


Approach

A Full-Stack Web Application.

Objectives

Frontend & Backend

**01**

Responsive design across devices.

02

Semantic structure and clear usability.

03

User authentication & database integration.

04

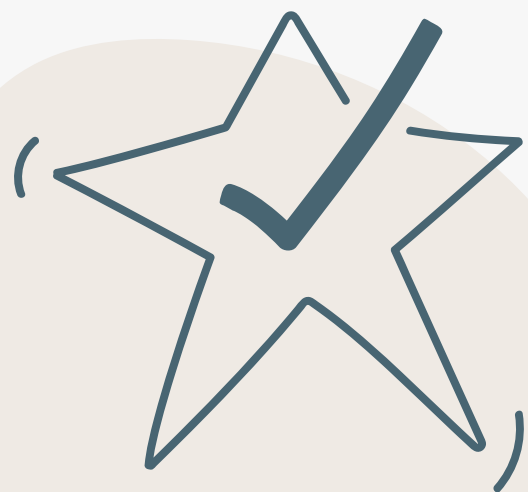
Personalized services (e.g., search, history, dashboard).



02

Project Goals and System Context

From a Static Site to a Full-Stack Web App



Project Goals & System Context



💡 Project Goal

- Help CS students with skill building
- Support master's prep & career planning
- Save programmes, jobs, and track browsing history
- Give personal help with saved items and history



🧱 Why Full-Stack?

- Static sites are read-only pages
- Users need accounts to save plans
- Need a database to store user data
- Need backend logic for real services



💻 Frontend

- Responsive design for all devices
 - Searchable content pages with clear navigation
 - Personal account area: view saved programmes, jobs, history
- Clean and user-friendly pages

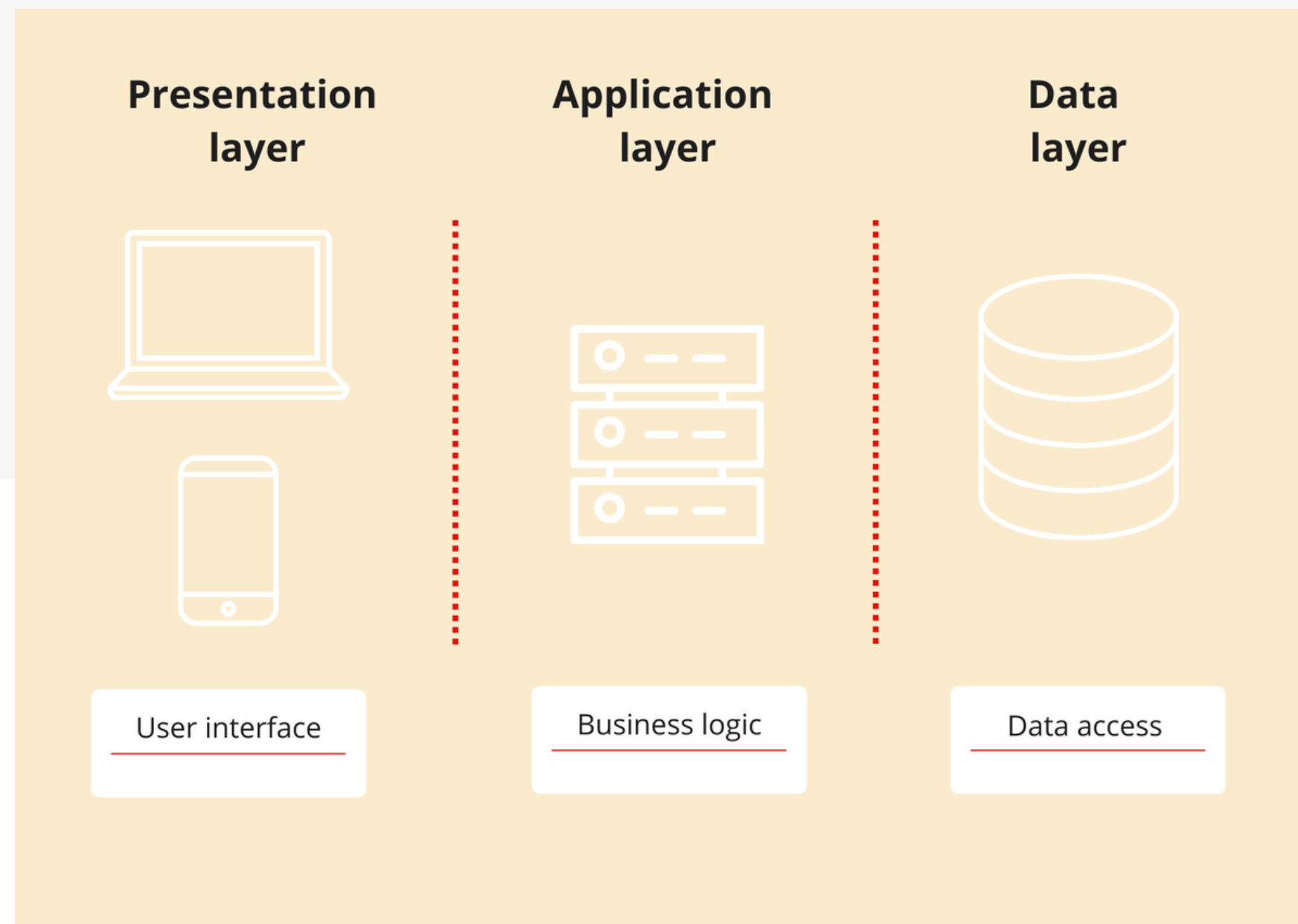


⚙️ Backend

- User sign-up and login
- Email verification and password reset
- Database stores profiles, saved items, and history
- Makes support personal, not generic



How Frontend and Backend Work Together



Frontend

- What users see and click
- Sends API requests to backend
- Shows results to users

Backend

- Receives user requests
- Checks user login session and permissions
- Talks to the database

Database

- Stores all user data
- Stores saved programmes, jobs, and browsing history
- Sends data back to backend



03

Core Services and Data Requirements

From static pages to data-driven

- System Service Architecture and Data Support Design



Overview of core service contents

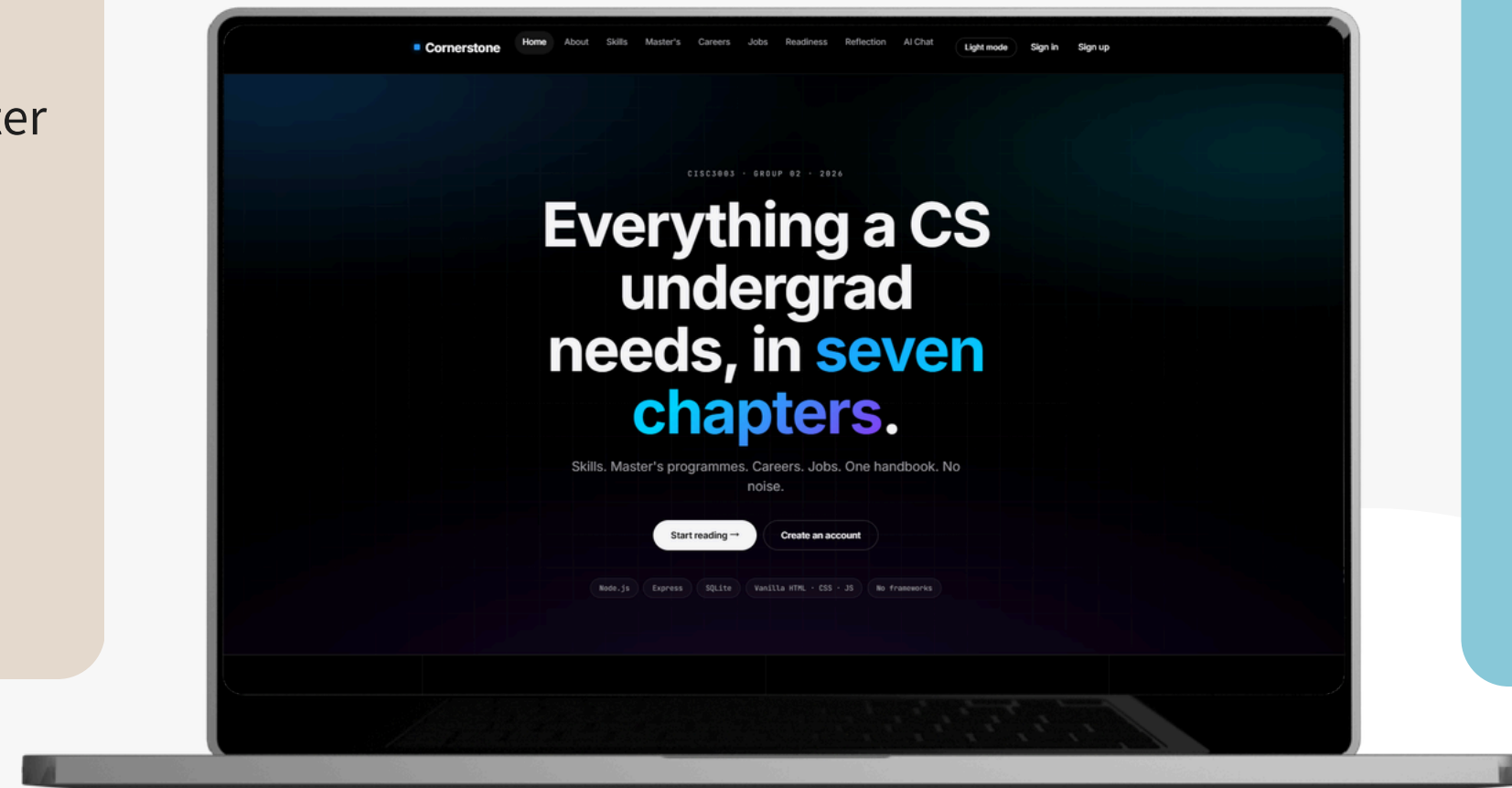
Our core service:

Providing precise information for the two career development paths of CS graduates



apply for postgraduate studies

- Actual ratings of global top computer universities and student evaluations
- Introduction and comparison of popular majors in each university
- Admission requirements and application suggestions for master's programs



prepare for direct employment

- Recommendations for mainstream technical positions (SDE, AI, data science, etc.)
- Job postings and requirements from renowned technology companies
- Skill matching suggestions and career development paths



Four core functional sections



skills development

- CoreRequirements: Programming languages, Data structures, Computer systems, Databases
- Advanced Specialization: Backend/Frontend/AI/System/Cloud Computing
- Soft Skills and Tools: Git, Technical Writing, English, Debugging Skills



Preparation for applying for further studies

- Application timeline and key tasks for freshmen to seniors
- Core preparation checklist: GPA, research/ internship, language exams, essays, recommendation letters
 - Overview of target universities' admission preferences and requirements



Career path

- Mainstream technical positions: Software development, AI/ML, Data science, SRE, Security, Mobile development, etc.
- Essential skill tags for each position
- Representative recruitment company examples (Google, Meta, OpenAI, ByteDance, etc.)



Institution rating

- Global computer science top universities data (MIT, Stanford, CMU, Oxford, ETH, etc.)
- Real student ratings and textual evaluations
- Customized filtering function by country, type, and rating range
- Users can submit ratings and evaluations to participate in community building



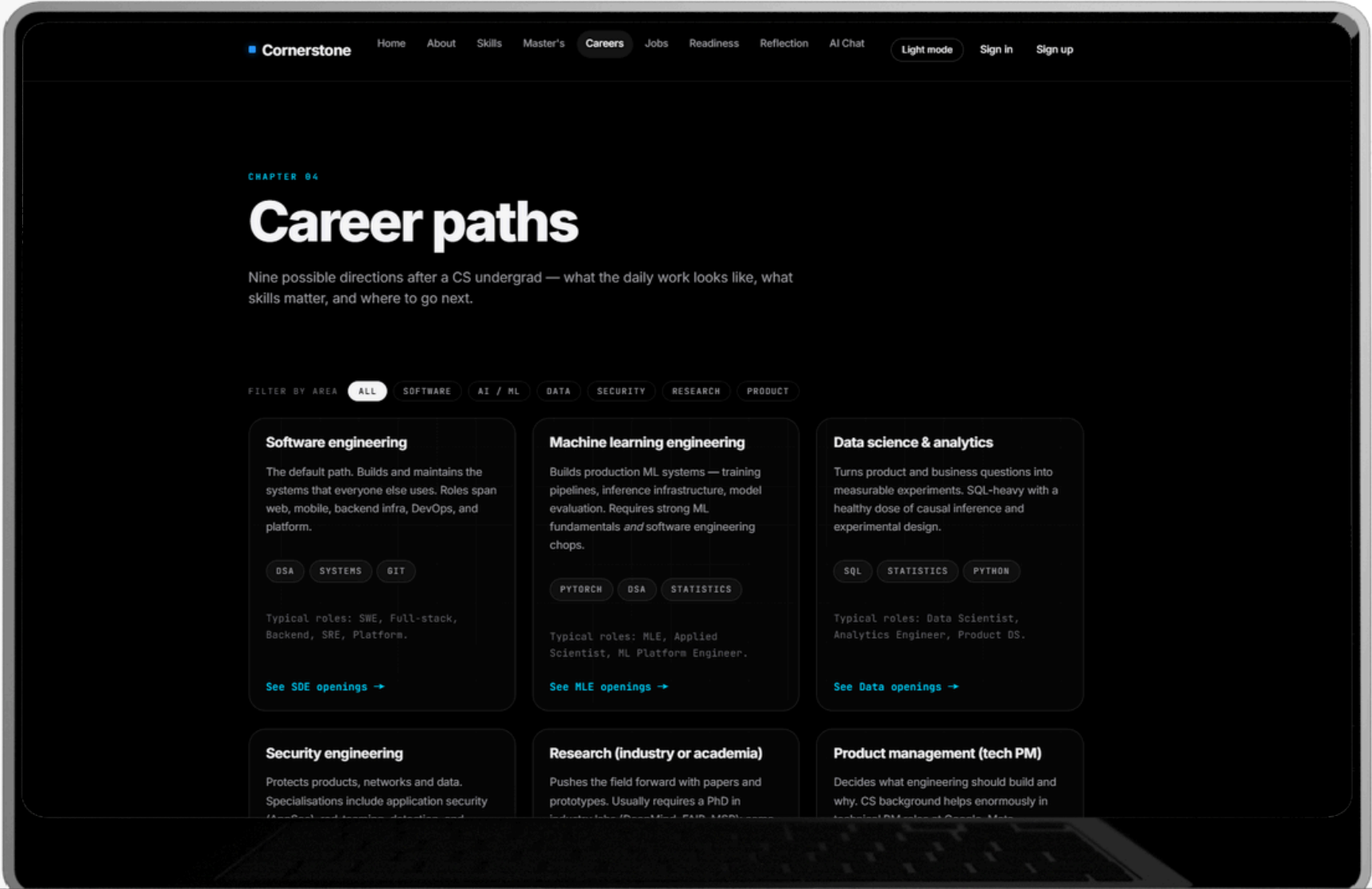
Detailed Explanation of the Career Path section



Career Paths section

This is the Career Paths section. We mapped out major directions after a CS undergrad, with 6 shown here: Software Engineering, Machine Learning, Data Science, Security, Research, and Product Management.

Each card contains three things: first, what the daily work looks like; second, key skill tags — for example, Software Engineering needs DSA, Systems; third, typical role titles. Users can also filter by area at the top, such as AI/ML or Data only.

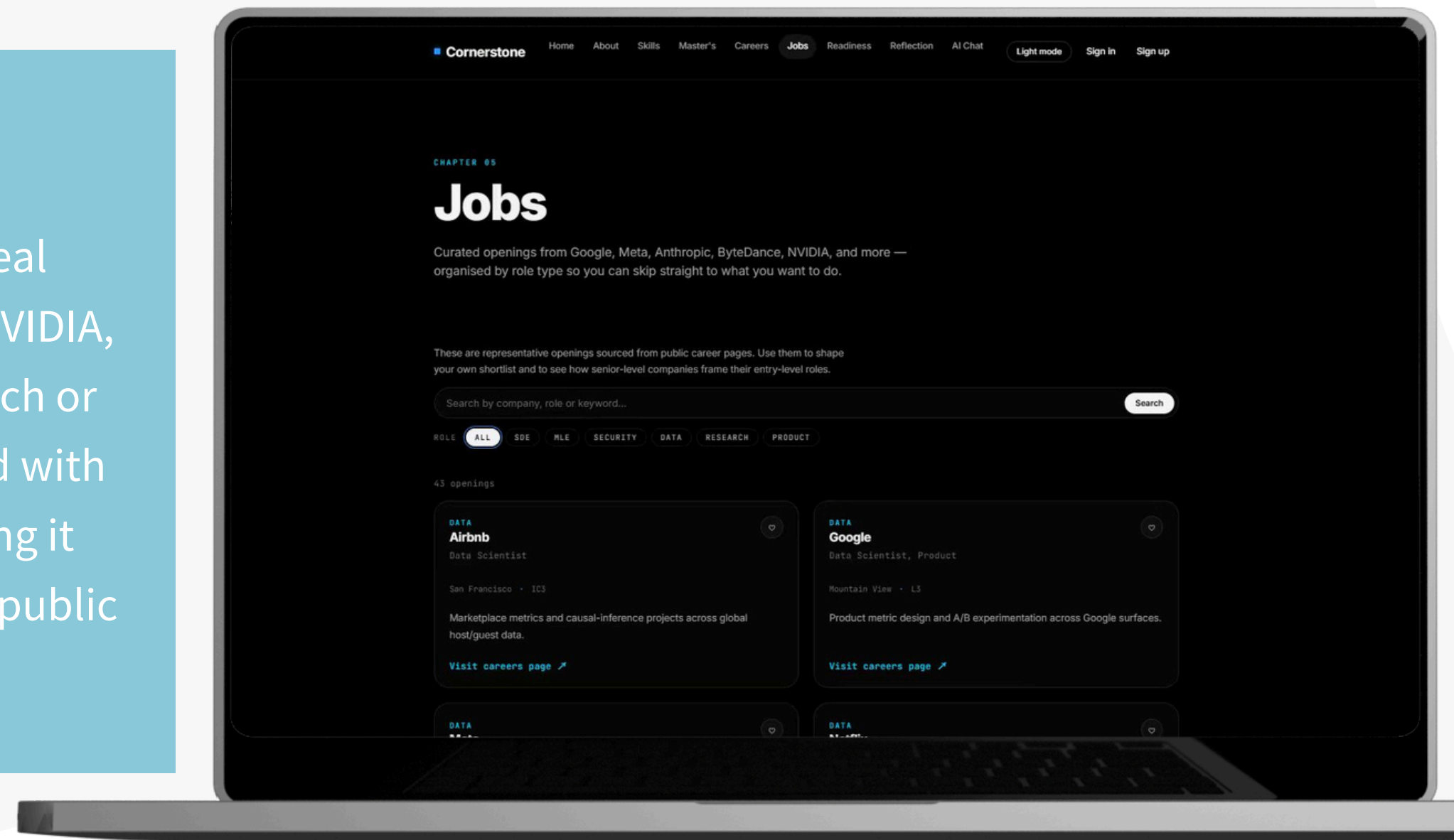


Detailed Explanation of the Jobs Section



Curated Job Openings

This is our Jobs page. We manually curated many real openings from top companies like Google, Meta, and NVIDIA, grouped by role — SDE, MLE, Data, etc. Users can search or filter by tags at the top, and each job is shown in a card with company, location, and a one-line description, making it easy to scan and shortlist. All listings are sourced from public career pages.



Data Requirements Summary



Overall Rank ▾ University ▾		Research & Discovery		Learning Experience	Employment
		Citations per Faculty ▾	Academic Reputation ▾	Faculty Student Ratio ▾	Employment Rate ▾
1	Massachusetts Institute of Technology (MIT) 📍 Cambridge, United States	100	100	100	100
2	Imperial College London 📍 London, United Kingdom <button>View Programmes</button>	95	99.6	99.3	100
3	Stanford University 📍 Stanford, United States	99.7	100	100	100
4	University of Oxford 📍 Oxford, United Kingdom	91	100	100	100
5	Harvard University 📍 Cambridge, United States	100	100	98.3	100

Information on overseas institutions

- The source of the crawl: U.S. News, QS official website, and pages of the CS departments of various universities
- Supporting functions: Basic data display and filtering on the institution'satingpage

Content of institution evaluation

- Crawled sources: Reddit, GradCafe, Zhihu
- Supporting function: Supplement the real word-of-mouth from the userevaluation community

Admission requirements for master's programs

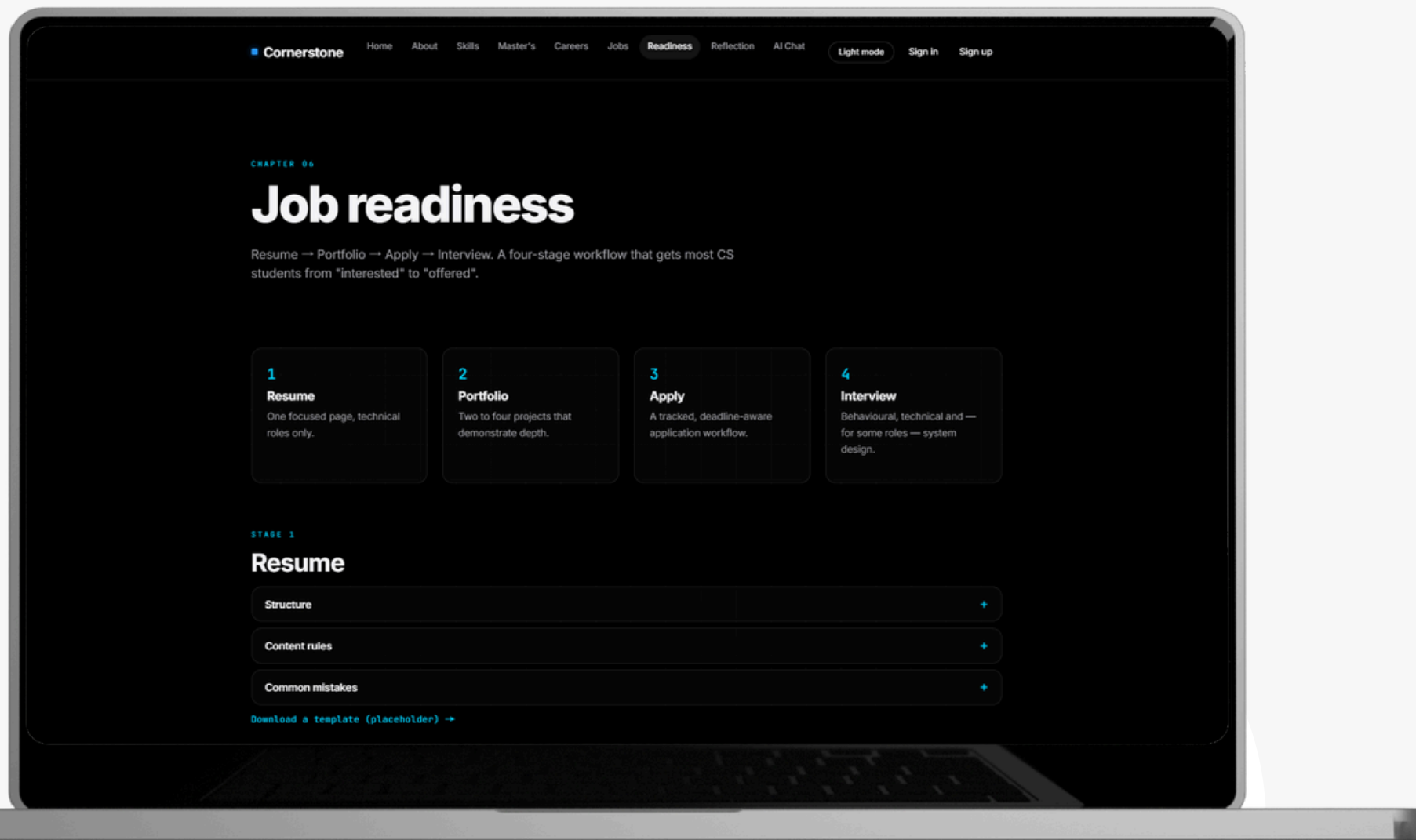
- Crawling sources: Official websites of graduate schools of variousuniversities, experience posts on application forums
- Supporting function: Reference for university admission preferences inthe master's preparation section

Technical position recruitment information

- Crawl sources: LinkedIn, Indeed, Glassdoor, and Careers pages of various companies
- Supporting functions: Job recommendations and company examples in the career path section



Web Development with AI Assistance



Web Design Refinement

- Used AI to optimize the visual style and layout, settling on a dark theme, card-based design, and chapter-based navigation.

Content Distillation

- Leveraged AI to condense large amounts of raw data (e.g., Master's programmes, job openings) into concise cards and one-line summaries for better readability.

AI Chat Feature

- Integrated an external API to add an AI Chat entry point for basic Q&A on career and school selection.



04

Project Tasks + Timeline

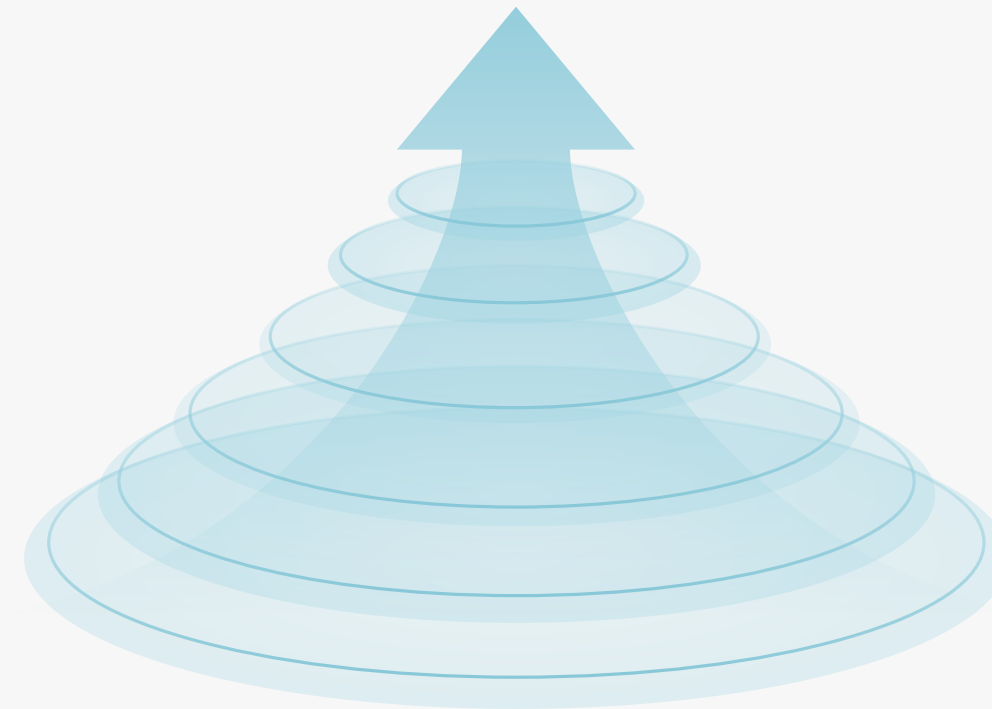
This section outlines the planned steps, expected deliverables, and timeline for our website project.

Core Goal



Clear Plan

Demonstrate that the team has a clear project plan



Structured Tasks

Show that tasks are structured and sequential

Transparent Deliverables

Provide transparency of deliverables at each stage



Project Tasks



Refine Project Scope – confirm content areas & target audience

Design Sitemap & Wireframe – plan site structure logically

Implement Frontend Pages – HTML/CSS/JS

Implement Backend Modules – server-side logic & APIs

Integrate Database & Services – connect frontend with backend

Test Responsiveness & Usability – desktop & mobile

Deploy the System – make website accessible online

Finalize PPT & Proposal – prepare submission materials



Four core functional sections



Day	Task
2026/4/17	Refine Project Scope & Define Audience
2026/4/17	Design Sitemap & Wireframe
2026/4/22	Implement Frontend Pages
2026/4/23	Implement Backend Modules & Integrate Database
2026/4/24	Test Responsiveness & Usability
2026/4/24	Deploy the System
2026/4/26	Finalize PPT & Proposal
2026/4/26	Buffer / Adjustments / Screenshots & Testing Results



Responsibilities for Part 4



Timeline Coordination: Ensure each task is completed on schedule, monitor overlapping steps.

Task Checklist & Progress Tracking: Keep daily records of progress to prevent delays.

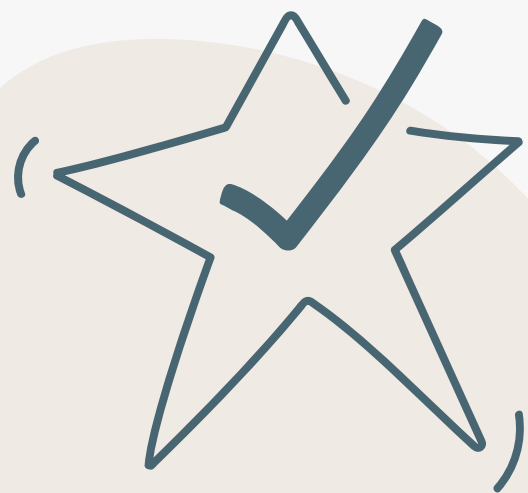
Deployment & Material Organization: Prepare hosting environment and collect all final files.

Final Submission Preparation: Consolidate website, PPT, proposal, and screenshots for submission.



05

Project Challenges and Evaluation



Four core challenges



From informational website to full-stack system

- Transitioning from a content-oriented website to a full-stack web application requires a significant expansion of project scope.
- The system must move beyond static information presentation and support dynamic user interaction, data processing, and service delivery.
- This challenge involves not only interface design, but also backend architecture, database connectivity, and functional service logic.



User data management

- The project involves multiple categories of user-related data, including registration details, verification records, password reset requests, search history, browsing history, and cart information.
- These data must be stored, updated, and retrieved in a structured and consistent manner.
- A further challenge is to organize the data model so that user activities can be reflected clearly in the dashboard and related service modules.



Frontend-backend integration

- Designing user interfaces is only one part of the system; the greater challenge lies in connecting frontend operations with backend processing in a coherent way.
- Core services such as user registration, login, search history, browsing history, dashboard updates, and cart actions all require complete data flow and interaction logic.
- The frontend must accurately send requests, while the backend must validate, process, and return the appropriate results.



Making the shopping cart meaningful

- Unlike conventional e-commerce platforms, our project does not center on direct retail transactions, which makes the design of the shopping cart less straightforward.
- The cart therefore needs to be reinterpreted as a resource and service selection mechanism suitable for an academic and career development platform.
- It may include selected guidance resources, planning services, saved materials, or personalized recommendation packages.



Criteria of Success



Part 1: Design, Accessibility & User Experience

- The website should present a clear, consistent, and professional interface that supports intuitive navigation and effective information delivery. (Nielsen Norman Group, 2017)
- The system should demonstrate responsive design, remaining readable and functional across desktop, tablet, and mobile devices. (Mozilla, 2025b)
- Page structure should follow semantic HTML (Mozilla, 2025a) and accessibility principles, including meaningful headings, readable text, and appropriate link and form design. (World Wide Web Consortium, 2025)



Part 2: Functional System Performance

- Complete full-stack system implementation
- Reliable signup, login, verification, and reset workflow
- Functional and personalized user dashboard
- Effective search, history, and cart services
- Stable frontend-backend integration



Part 3: Relevance, Deployment & Project Value

- Content aligned with CS students' needs
- Meaningful and practical project-specific services
- Successful deployment with public Web URL
- Consistency between proposal and final implementation
- Clear academic, technical, and user-oriented value

Evaluation methods: functionality and responsiveness



Evaluation checklist

- Verification of all required system modules, including signup, login, password reset, dashboard, search, history, and cart functions
- Confirmation that frontend pages, backend services, and database interactions operate as intended
- Use of a structured checklist to ensure consistency between project requirements and implementation outcomes
- Identification of incomplete features, logic errors, and revision priorities before final deployment

Cross-device testing

- Examination of the system across desktop, tablet, and mobile screen sizes to ensure responsive presentation (Mozilla, 2025b)
- Testing of layout stability, readability, button accessibility, and navigation clarity under different display conditions
- Review of browser compatibility to reduce interface inconsistencies across common platforms
- Validation that essential user functions remain usable and visually coherent on multiple devices





Evaluation methods: functionality and responsiveness

Workflow testing

- Simulation of complete user journeys, such as registration, verification, login, searching, viewing history, and managing cart items
- Assessment of whether system operations follow a logical and uninterrupted sequence from the user's perspective
- Detection of broken transitions, missing feedback, or data flow issues between frontend and backend processes
- Evaluation of whether the platform supports realistic and meaningful interaction scenarios for student users

Usability review

- Review of navigation clarity, information hierarchy, and readability throughout the website (World Wide Web Consortium, n.d.)
- Assessment of whether users can understand page purposes and complete tasks without unnecessary difficulty
- Examination of form design, button labeling, content organization, and overall interface consistency (World Wide Web Consortium, 2025)
- Determination of whether the final system remains aligned with the needs of computer science students and the original project goals



06

Team Structure & Project Learning



team structure



Frontend----Luo Yiming, Yang

UI/UX design, 16 HTML pages, responsive layout, dark/light theme, all client-side interactions

Data, AI & DevOps----Zhang Zhexuan, Chen Siyi

Data collection (programmes, jobs), AI chat backend (DashScope API integration), deployment & server config



Backend & debug----Yang Yujie, Zhang Jieding, Jiao Zinan

Core backend modules (auth, API routes, database), testing, debugging & stability assurance

outcomes and challenges



Technical Growth

- Static to Dynamic: Evolved from basic HTML/CSS to PHP-MySQL integration, implementing user login and data persistence
- PHP Session Management: Learned to use \$_SESSION for login states and \$_COOKIE for "remember me" features (implemented by Group 1)
- Database Design: Mastered MySQL table design and PHP MySQLi connection (Group 1), while Group 2 learned to structure HTML content for database storage
- Responsive Design: Group 2 implemented CSS Media Queries for mobile/tablet/desktop without external frameworks



Collaboration Insights

- • Application timeline and key tasks for freshmen to seniors
- • Core preparation checklist: GPA, research/ internship, language exams, essays, recommendation letters
- • Overview of target universities' admission preferences and requirements





PROJECT RELATED REFERENCE



Mozilla. (2025a, November 4). *HTML: A good basis for accessibility*. MDN Web Docs. <https://developer.mozilla.org/en-US/docs/Learn/Accessibility/HTML>

Mozilla. (2025b, December 19). *Responsive web design*. MDN Web Docs. https://developer.mozilla.org/en-US/docs/Learn_web_development/Core/CSS_layout/Responsive_Design

Nielsen Norman Group. (2017, October 1). *First impressions matter: How designers can support users' automatic cognition*. <https://www.nngroup.com/articles/first-impressions/>

World Wide Web Consortium. (n.d.). *Introduction to web accessibility*. W3C Web Accessibility Initiative. <https://www.w3.org/WAI/fundamentals/accessibility-intro/>

World Wide Web Consortium. (2025, May 6). *Web Content Accessibility Guidelines (WCAG)*. <https://www.w3.org/WAI/WCAG21/quickref/>





Thank you

Thank you for watching

